

6th Grade Accelerated Unit 12 Assessment Guide

General Information	The assessment covers two benchmarks, MA.7.AR.3.1 and MA.7.AR.3.2. Encouraging students to correctly show units during instruction is important but this assessment does not rely on the student providing the unit as the units are provided in the table or answer space. Teachers may choose to edit the assessment to require students to give the correct unit. A common error that students make is an incorrect set-up of a proportion. If students do this throughout the questions assessing MA.7.AR.3.2, consider working with these students in a small group to determine the reason for their misconception and to provide the student a chance to show mastery after remediation. Sample work for the answers is provided but may not be the way students show their work. It is provided to support teachers. The benchmark MA.7.AR.3.1 covers several different types of questions that involve percentages. An assessment that would cover the variety implied by the benchmark would be too long for a class period. Teachers may wish to use other Math Nation tools such as Check Your Understanding or Test Yourself to further gauge student understanding on specific types of questions. If teachers find that students need support, then Stepping Stones questions may provide students with additional scaffolding. Teachers will find that the students may benefit from occasional review of this standard. As students continue through the 6th grade accelerated content, their fluency with rational numbers and proportional relationships should improve, so an intentional spiral review on common errors or common questions may help students strengthen their understanding. At the beginning of the assessment is a box with the statements: <i>Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit, show the operation that you entered into the calculator and the result. It is helpful if you label the operation with information from the word problem.</i> This suggestion can also help a student self-scaffold a problem. As you consider the point distribution, consider the sub-steps students may take. Consider that not every student needs to show sub-steps, so your point distribution should not punish students who can think through the problem and find an answer using a calculator such as the Desmos scientific calculator. It is possible that students can enter these problems as one expression into a calculator such as this.
Calculator Information	The questions on this assessment are calculator.

Reference Sheet	Students should have access to the conversions on the Grade 7 B.E.S.T. Mathematics Reference Sheet.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1	Benchmark(s)	MA.7.AR.3.2
	$100000 \text{ km} \cdot \frac{1 \text{ mile}}{1.61 \text{ km}} \approx 62111.8$ OR $100000 \text{ km} \cdot \frac{0.62 \text{ mile}}{1 \text{ km}} = 62000$	Recommended Scoring (14 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider giving students 6 points for the work shown and 8 points for the correct answer.
Question	2	Benchmark(s)	MA.7.AR.3.2
	$\frac{1}{5} = \frac{x \text{ ounces of sugar}}{27 \text{ ounces}}$ 5.4 ounces of sugar	Recommended Scoring (14 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider giving students 6 points for the work shown and 8 points for the correct answer.
		Additional Notes	Students may choose to solve the problem using a variety of methods that indicate an understanding of the proportional relationship.

Question	3	Benchmark(s)	MA.7.AR.3.2
$\frac{56 \text{ days}}{500 \text{ miles}} = \frac{x \text{ days traveling along the river}}{25.3 \text{ miles}}$ 2.8 days traveling along the river		Recommended Scoring (14 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider giving students 6 points for the work shown and 8 points for the correct answer.
Question	4	Benchmark(s)	MA.7.AR.3.1
\$133.50		Recommended Scoring (12 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider giving students 6 points for determining the price paid for one adult and 6 points for the correct answer.
Question	5	Benchmark(s)	MA.7.AR.3.1
\$8,603.60		Recommended Scoring (12 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider giving students 6 points for determining the interest after 8 years and 6 points for the correct answer.

Question	6	Benchmark(s)	MA.7.AR.3.1
$9,787$		Recommended Scoring (12 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	7	Benchmark(s)	MA.7.AR.3.1
$\frac{282 - 279.2}{279.2} \approx 0.010028 \dots$ $\frac{282 - 285}{285} \approx 0.010526 \dots$ $\frac{282 - 283.06}{283.06} \approx -0.003744 \dots$ Complete the statement. Tool(s) ☐ A only ☐ B only ☐ C only ☐ A and B ☒ A and C ☐ B and C ☐ A, B, and C ☒ do(es) ☐ do(es) not fit within the accepted percent error.		Recommended Scoring (22 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider giving students 6 points for the work for determining whether each tool fits in the error and 4 points for a correct statement.

Students could also correctly complete the statement with, "Tool(s) B only do(es) not..."